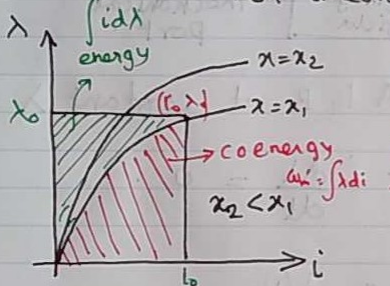
Lecture 21

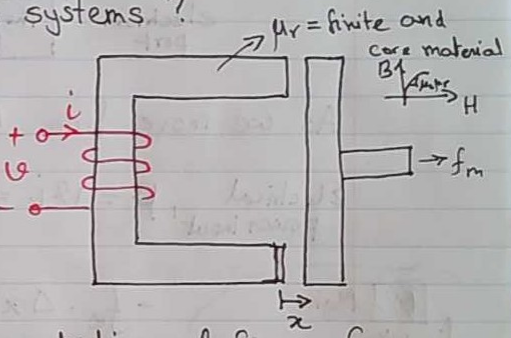
Last time : Q. When can a system become "electrically" nonlinear? $\lambda = L(x,i)\, i$

Q. How can we understand system's energy and co-energy, if the system is electrically non-linear ?

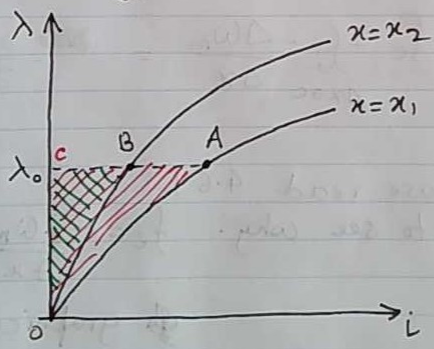
Graphical analysis in i-λ plane

Q. Is our approach of force computation valid for electrically non-linear systems ?





Q. What does it mean – computation of force from the energy perspective ?



$x_1 > x_2$

Energy, $W_m = f(\lambda, x)$

Energy, when the plunger $x = x_1$,

$$W_m(\lambda_0, x_1) = \int_0^{\lambda_0} i\, d\lambda \quad (x = x_1)$$

$$= \text{OAC}$$

Energy, when the plunger $x = x_2$,

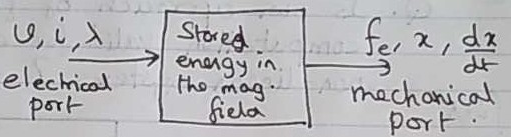
$$W_m(\lambda_0, x_2) = \int_0^{\lambda_0} i\, d\lambda = \Delta OBC$$

∴ What is our change in energy ?

$\Delta W_m = \Delta OCB - \Delta OAB$. (negative)

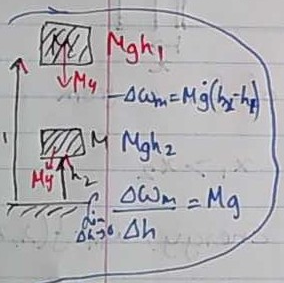
$= -\Delta OAB$

$\Delta x = x_2 - x_1$ (negative)



As we move from A to B at constant λ.

Electrical power input, $P = vi = i\frac{d\lambda}{dt} = 0$.



Analogy.

$-f_e \cdot \Delta x = \Delta W_m$

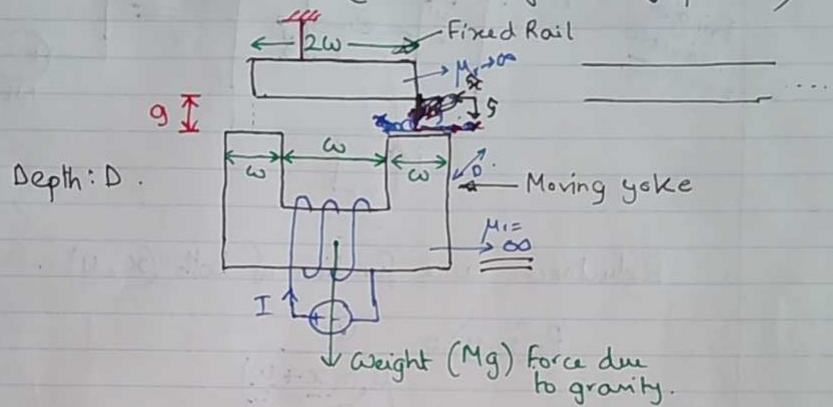
work done. change in energy.

$f_e = \lim_{\Delta x \to 0} -\frac{\Delta W_m}{\Delta x} = -\frac{\partial W_m}{\partial x}\Big|_{\lambda = \text{const}}$.

Please read 4.6.1 to see why. $f_e = -\frac{\partial W_m}{\partial x}\Big|_{\lambda=\text{const}} = \frac{\partial W_m'}{\partial x}\Big|_{i=\text{const}}$

graphically.

Real-world example: Magnetic levitation (Electromagnetic suspension)



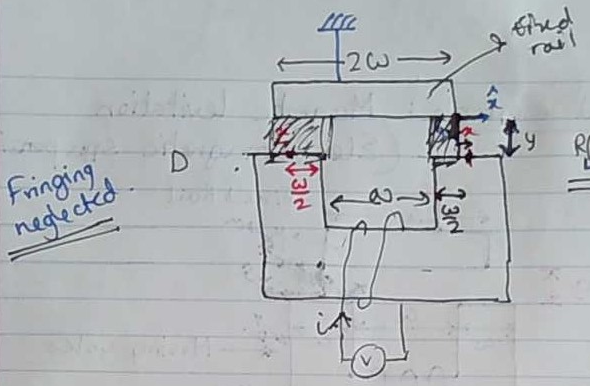
Q. The moving yoke can move both in x-direction and y direction — Multiport systems.

↓ two mechanical degree of freedom (x, y)

→ One electrical port.

Of course, we have calculated f_y many times. except that now, the fixed rail does not completely overlap with the yoke width 'w'.

In order to understand, the lateral force, let us perturb the yoke laterally.



Reluctances are function of both (x, y).

$$R_L(x,y) = \frac{y}{\mu_0 D\left(x + \frac{\omega}{2}\right)}$$

$$R_R(x,y) = \frac{y}{\mu_0 D\left(\frac{\omega}{2} - x\right)}$$

Now, we can write

$$d\omega_m' = \lambda\, di + f_{e,x}\, dx + f_{e,y}\, dy.$$

two mechanical "port"/degree of freedom.

$$\therefore \quad \omega_m' = \int_0^i \lambda\, di' \quad \text{while } x \text{ and } y \text{ are constant}$$

$$= \frac{1}{2} L(x,y)\, i^2 = \frac{1}{2} \frac{N^2}{\frac{y}{D\mu_0\left(x+\frac{\omega}{2}\right)} + \frac{y}{D\mu_0\left(\frac{\omega}{2}-x\right)}}$$

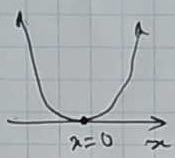
$f_e|_{x\text{-direction}} = \frac{i^2}{2}\frac{\partial L}{\partial x}$

$$= \frac{1}{2} \frac{\mu_0 D N^2}{\frac{y}{\frac{\omega}{2}+x} + \frac{y}{\frac{\omega}{2}-x}} = \frac{1}{2}\mu_0 \frac{DN^2}{y\omega}\left(\frac{\omega^2}{4} - x^2\right) i^2$$

$$\therefore \quad f_e|_{y\text{-direction}} = \frac{\partial L(x,y)}{\partial y}\bigg|_{\substack{i,x \\ =\text{const.}}} = -\frac{1}{4}\frac{D\mu_0 N^2 i^2}{y^2 \omega}$$

$$f_e|_{x\text{-direction}} = \frac{i^2}{2}\frac{\partial L(x,y)}{\partial x}$$

i = const
y const.



$$= \frac{i^2}{2}\frac{\mu_0 D N^2}{w y}\cdot\left[-2x\right]$$

$$= \frac{-2\mu_0 D N^2 i^2 x}{w y}$$

What does it mean?

* Control current such that

$$f_e|_{y\text{-direction}} = Mg$$

(M: mass, g: ms)

Two electrical port + 1 mechanical port.